PARSA NIAVAND

AI Engineer | **Machine Learning / LLM Engineer**

Python / PyTorch / LLMs / RAG Architectures / Deep Learning

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Software and AI engineer with 3 years production engineering experience, combining deep learning fundamentals (CNNs, VAEs) with practical LLM/RAG pipeline design, vector search, and scalable backend/API deployment.

PROFESSIONAL EXPERIENCE

Software Engineer Codesheen

Full-time, On-site Oct 2023 – Jul 2026

- Worked as one of 4 core engineers (alongside 2 designers and 2-3 remote contributors), delivering on **8+ concurrent client and internal production web systems**.
- Built full-stack features using Next.js, React, Node.js, and NestJS, implementing clean APIs and PostgreSQL schemas.
- Maintained shared monorepos (Turborepo) with end-to-end type safety, improving team velocity and **cutting build times by 30%**.
- Optimized frontend asset delivery, Core Web Vitals, and backend caching layers to maintain sub-second page loads.

Selected Team Project Deliverables:

Goldenbat GPS Fleet Tracking Sep 2024 – Jan 2025 Live Demo | Dashboard

Co-built a real-time fleet tracking platform handling **10,000+ coordinate updates/min**. Developed Next.js mobile PWA and contributed to GIS mapping dashboard with live tracking, geofencing, and user troubleshooting.

Battery ESP/MSK E-Commerce Platform Dec 2025 – Jun 2026 Live Demo

Shipped two production bilingual e-commerce sites and an admin panel in a shared monorepo; translated Figma specs into responsive React UI and engineered warranty key verification logic.

Taxiland Ride-Sharing Platform May 2024 – Aug 2024 Archived MVP

Contributed to passenger/driver mobile interfaces and built backend dispatch endpoints for driver assignment and support tickets.

AI & PERSONAL PROJECTS

RAG Pipeline & Vector Search Experiments Personal Project AI / LLMs

Built RAG workflows using Python, vector embeddings, document chunking, and similarity search for querying unstructured document datasets.

Deep Learning Anomaly Detection for Mineral Zones Nov 2025 GitHub

Developed unsupervised deep learning models for geological anomaly and mineral zone detection using Variational Autoencoder (VAE) and Autoencoder architectures, identifying prospective zones via latent reconstruction error.

Cat Emotion Recognition (Computer Vision CNN) Aug 2025 GitHub

Trained and tuned a Convolutional Neural Network (CNN) in PyTorch to classify multi-class emotion patterns from facial features, utilizing data augmentation and transfer learning to reach **70% classification accuracy**.

Atrafian Social Platform (On-Device NSFW Filtering) Jan 2026 – Mar 2026 Landing Demo | Dashboard

Implemented client-side on-device image moderation pipeline to pre-screen media uploads for NSFW content before server ingestion, eliminating backend compute costs while preserving user privacy.

SKILLS

AI & Machine Learning: PyTorch, Scikit-learn, CNNs, VAE / Autoencoders, Computer Vision, NumPy, Pandas.

LLMs & Generative AI: RAG Pipelines, Vector Search, Semantic Chunking, Embeddings, Prompt Engineering, LangChain, API Integration.

Data & Vector Stores: PostgreSQL, pgvector, Redis, ChromaDB, Elasticsearch, MinIO, SQL, Data Pipelines.

Languages: Python, TypeScript, JavaScript, SQL, Bash, HTML5/CSS3.

Backend & Deployment: FastAPI, Node.js, NestJS, Next.js, REST APIs, WebSockets, Docker, Linux, Git.

Engineering Practices: Model Evaluation, CI/CD, Unit Testing, Monorepos, Agile Collaboration.

EDUCATION

B.S. in Computer Science Graduated 2026 Science and Research Branch (SRBIAU)